

06/05/17(E)

K. J. Somaiya College of Engineering, Mumbai-77
(Autonomous College Affiliated to University of Mumbai)

End Semester Exam
April - May 2017

Max. Marks: 100

Class: SYBTech

Name of the Course: Analysis of Algorithms

Course Code: UITC403

Duration: 03 Hrs.

Semester: IV

Branch: IT

Instructions:

- (1) All Questions are Compulsory
- (2) Draw neat diagrams
- (3) Assume suitable data if necessary

Question No.		Max. Marks																										
Q 1 (a)	Explain the terms space complexity and Time complexity, with suitable example. Analyze the following algorithm to find time complexity in terms of Big-Oh. <i>Algorithm prefixAverages (X) :</i> <i>Input:</i> An n-element array X of numbers. <i>Output:</i> An n-element array A of numbers such that $A[i]$ is the average of elements $X[0], \dots, X[i]$. <i>Let A be an array of n numbers.</i> $s \leftarrow 0$ <i>for $i \leftarrow 0$ to $n-1$ do</i> $s \leftarrow s + X[i]$ $A[i] \leftarrow s / (i+1)$ <i>return array A</i> OR Explain the RAM model. Discuss the need of analysis of algorithms. Write a pseudo-code for finding substring in a given string and perform running time analysis of the same.	10																										
Q 1 (b)	Explain steps in substitution method for solving recurrences. Use recursion tree method to determine a good asymptotic upper bound on the recurrence $T(n) = 3T(n/2) + n.$ Use the substitution method to verify the answer.	10																										
Q2 (a)	Following has been the temperature (in degree Celsius) of cities in Maharashtra. Arrange and list the cities in decreasing order of temperature (in degree Celsius) using Shell sort. Provide pseudo-code for the same. <table><tr><td>Temp (in degree Celcius)</td><td>40</td><td>36</td><td>42</td><td>45</td><td>37</td><td>44</td><td>29</td><td>39</td><td>28</td><td>27</td><td>30</td><td>26</td></tr><tr><td>City</td><td>Auranga</td><td>Beed</td><td>Chandra</td><td>Dhule</td><td>Gondia</td><td>Jalgaon</td><td>Kolhapur</td><td>Latur</td><td>Mumbai</td><td>Pune</td><td>Raigad</td><td>Sangli</td></tr></table> OR Arrange and list the cities given above in increasing order of temperature (in degree Celsius) using Heap sort. Provide pseudo-code for the same.	Temp (in degree Celcius)	40	36	42	45	37	44	29	39	28	27	30	26	City	Auranga	Beed	Chandra	Dhule	Gondia	Jalgaon	Kolhapur	Latur	Mumbai	Pune	Raigad	Sangli	12
Temp (in degree Celcius)	40	36	42	45	37	44	29	39	28	27	30	26																
City	Auranga	Beed	Chandra	Dhule	Gondia	Jalgaon	Kolhapur	Latur	Mumbai	Pune	Raigad	Sangli																

Q2 (b)	Name the efficient algorithm for sorting provided data. Justify how it is efficient. Sort the numbers with the said algorithm (Ascending order) Data : 531, 317, 910, 199, 345, 7456, 3210, 654, 433, 532	08																																				
Q3 (a)	Given is the list of students and their marks, arrange students in increasing order of their marks using Merge sort. Write the technique being used and explain how it is applied sorting. <table><tr><td>Marks</td><td>70</td><td>75</td><td>66</td><td>60</td><td>67</td><td>72</td><td>82</td><td>79</td><td>78</td><td>80</td><td>59</td><td>55</td></tr><tr><td>Student</td><td>Dipen</td><td>Jay</td><td>Aditya</td><td>Mihir</td><td>Saurabh</td><td>Jigar</td><td>Viraj</td><td>Hiral</td><td>Shreya</td><td>Aman</td><td>Chirag</td><td>Arvind</td></tr></table> OR Arrange the students given above in decreasing order of their marks using Quick sort. Provide the related pseudo-code.	Marks	70	75	66	60	67	72	82	79	78	80	59	55	Student	Dipen	Jay	Aditya	Mihir	Saurabh	Jigar	Viraj	Hiral	Shreya	Aman	Chirag	Arvind	12										
Marks	70	75	66	60	67	72	82	79	78	80	59	55																										
Student	Dipen	Jay	Aditya	Mihir	Saurabh	Jigar	Viraj	Hiral	Shreya	Aman	Chirag	Arvind																										
Q3 (b)	Prince, the ^{only} jewel thief wants to rob a jewelers shop. He can carry ^{of} only weight up to 8kg without getting caught. Help him choose the box ^{of} jewels so as to have a most profitable robbery. The weight and profit of the jewel boxes is as follows. (Boxes are packed and it will be risky to open during stealing them) <table><tr><td>Box of Jewel</td><td>Pearl</td><td>Ruby</td><td>Topaz</td><td>Emerald</td><td>Diamond</td></tr><tr><td>Weight (in kg.)</td><td>4</td><td>3</td><td>5</td><td>6</td><td>2</td></tr><tr><td>Profit</td><td>9</td><td>7</td><td>10</td><td>9</td><td>3</td></tr></table>	Box of Jewel	Pearl	Ruby	Topaz	Emerald	Diamond	Weight (in kg.)	4	3	5	6	2	Profit	9	7	10	9	3	08																		
Box of Jewel	Pearl	Ruby	Topaz	Emerald	Diamond																																	
Weight (in kg.)	4	3	5	6	2																																	
Profit	9	7	10	9	3																																	
Q4 (a)	<i>optimal</i> Bagginses of The Shire loves travelling. Help Bilbo Baggins plan his ^{optimal} journey from, and back to, his native-The Shire using Branch and Bound. The distances between the places are as follows <table><tr><td></td><td>The Shire</td><td>Bree</td><td>Gondor</td><td>Mirkwood</td><td>Rohan</td></tr><tr><td>The Shire</td><td>-</td><td>10</td><td>8</td><td>9</td><td>7</td></tr><tr><td>Bree</td><td>10</td><td>-</td><td>10</td><td>5</td><td>6</td></tr><tr><td>Gondor</td><td>8</td><td>10</td><td>-</td><td>8</td><td>9</td></tr><tr><td>Mirkwood</td><td>9</td><td>5</td><td>8</td><td>-</td><td>6</td></tr><tr><td>Rohan</td><td>7</td><td>6</td><td>9</td><td>6</td><td>-</td></tr></table>		The Shire	Bree	Gondor	Mirkwood	Rohan	The Shire	-	10	8	9	7	Bree	10	-	10	5	6	Gondor	8	10	-	8	9	Mirkwood	9	5	8	-	6	Rohan	7	6	9	6	-	12
	The Shire	Bree	Gondor	Mirkwood	Rohan																																	
The Shire	-	10	8	9	7																																	
Bree	10	-	10	5	6																																	
Gondor	8	10	-	8	9																																	
Mirkwood	9	5	8	-	6																																	
Rohan	7	6	9	6	-																																	
Q4 (b)	Discuss how Backtracking works for N-Queen Problem. Write pseudo-code for the same. OR Discuss how Backtracking works for Sum of subsets Problem with a suitable example. Write pseudo-code for the same.	08																																				
Q5 (a)	Write pseudo code for Robin Karp algorithm. Explain with a suitable example.	08																																				
Q5 (b)	Find longest Common Subsequence of strings "THEFELLOWSHIP OF THE RING" and "SOMAIYAENGINEERING"	12																																				